Listing 2.15 Registering middleware with an extension method

```
namespace WebApplication1
{
    public static class Extensions
    {
        public static IApplicationBuilder UseIpAddressMiddleware(this
IApplicationBuilder app)
        {
            return app.UseMiddleware<IpAddressMiddleware>();
        }
    }
}
```

Then the extension method is used in the same way as all the other extension methods you have encountered for registering framework middleware:

```
app.UseIpAddressMiddleware();
```

In this case, you will probably want to register this middleware after the static files middleware, so it doesn't log the IP address for the same visitor for every requested file.

Middleware that follows the convention-based approach is created as a *singleton* when the application first starts up, which means there is only one instance created for the lifetime of the application. This instance is reused for every request that reaches it.

IMPLEMENTING MIDDLEWARE

The recommended approach to writing new middleware classes involves implementing the `IMiddleware` interface, which exposes one method:

```
Task InvokeAsync(HttpContext context, RequestDelegate next)
```